1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights tumors with TGF-β–driven fibrosis and GREM1/BMP-antagonism that are not being captured by current prognostic stratification — why does a fibrotic, pro■metastatic microenvironment fail to produce a consistent signal for progression across patients? 2) Tier 1 paths used: (a) TGF-β → fibrogenic, immunosuppressive stroma (core link to tissue remodeling); (b) GREM1 → BMP antagonism associated with lymph■node metastasis (core link to metastatic potential). I built on these because they provide a mechanistic axis connecting stromal remodeling to enhanced dissemination. 3) Tier 2 evidence supporting the claim: MSH6 introduces a genomic■instability/immunogenicity modifier that could interact with stromal effects; PSA/pro■PSA and SARIFA point to cross■modal (biochemical/imaging/histologic) signals that can complement stromal readouts; SDC2/TFPI2 methylation suggests an orthogonal, noninvasive epigenetic detection route. These support testing whether microenvironmental imaging features plus molecular signals improve prognostication. 4) Why PFI as the endpoint: GREM1's link to lymph■node metastasis and TGF■β's role in invasion imply an effect on recurrence/progression timing more directly than on long-term OS (which is confounded by subsequent therapies); PFI is therefore the most sensitive clinical endpoint to detect effects of pro■metastatic microenvironmental programs in available cohorts. 5) Deliberate exclusions: I excluded cross■disease grading constructs (e.g., Gleason) and broad, multi■gene lists from the core hypothesis to preserve mechanistic clarity and meet the 1 imaging + ≤2 gene constraint. Epigenetic markers (SDC2/TFPI2) and genomic modifiers (MSH6), as well as PSA/pro■PSA and SARIFA, are relegated to secondary_modifiers for follow■up analyses rather than the primary mechanistic claim?

Working Hypothesis

High stromal-to-tumor ratio (TSR) indicates a TGFB1-driven fibrotic microenvironment that cooperates with GREM1-mediated BMP antagonism to establish a pro-metastatic niche facilitating early dissemination. In breast cancer, tumors with high TSR alongside elevated TGFB1 and GREM1 expression will show shorter progression-free interval, and the TSR–TGFB1–GREM1 signature will provide additive prognostic value beyond TSR alone.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGFB1, GREM1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

GREM1: GREM1/BMP-antagonism that are not being captured by current prognostic stratification — why does a fibrotic, pro■metastatic microenvironment fail to produce a consistent signal for progression across patients? 2) Tier 1 paths used: (a) TGF-β → fibrogenic, immunosuppressive stroma (core link to tissue remodeling); (b) GREM1 → BMP antagonism associated with lymph■node metastasis (core link to metastatic potential).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	MSH6	Supporting modifier from knowledge graph
Secondary	PSA	Supporting modifier from knowledge graph
Secondary	SARIFA-positivity	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective cohort analysis across breast cancer datasets with quantified TSR from digital pathology, TGFB1 and GREM1 expression data, and reported PFI; use multivariable Cox models with interaction terms to test whether TSR combined with TGFB1 and GREM1 improves prognostic performance over TSR alone.

Prediction 2: In vitro 3D co-culture systems pairing breast cancer cells with fibroblasts engineered to overexpress TGFB1 and GREM1; assess ECM remodeling, invasion, and response to TGF-β/BMP pathway modulators.

Prediction 3: In vivo orthotopic breast cancer models to evaluate metastasis in relation to TSR, TGFB1, and GREM1 status and to test the effect of pathway inhibitors on nodal/hematogenous spread.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SDC2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TFPI2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.